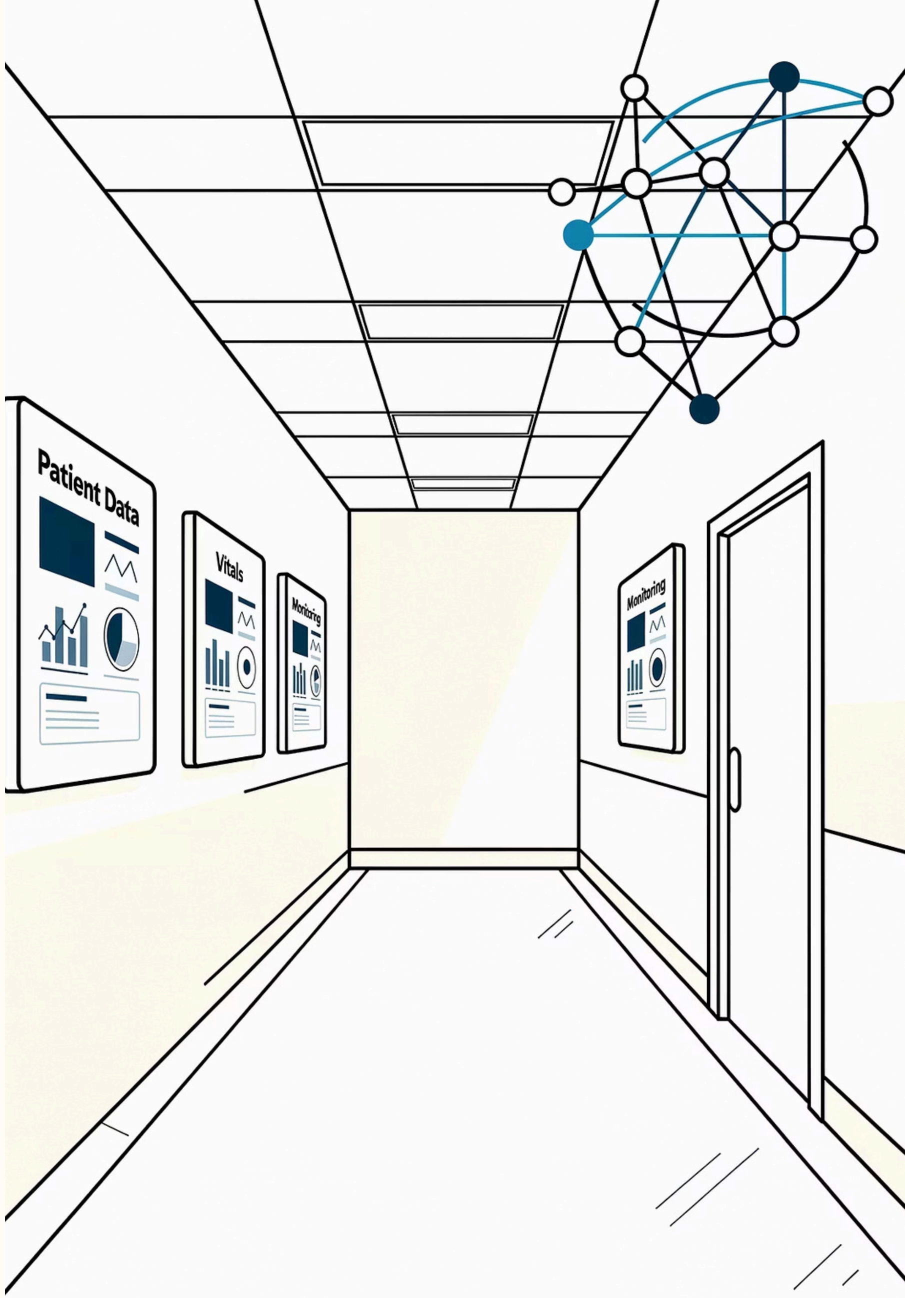
Bridging the Gaps: Healthcare Interoperability Standards in Canada and Beyond

A comprehensive exploration of the standards, frameworks, and initiatives shaping the future of health data exchange — from pan-Canadian blueprints to globally recognized protocols like HL7 FHIR.

HEALTHCARE POLICY DIGITAL HEALTH INTEROPERABILITY



The Challenge: A Fragmented Health Data Landscape

Canada's healthcare system, like those of many nations, is built on decades of independently developed information systems — each with its own data formats, terminology standards, and technical architectures. The result is a deeply fragmented landscape where critical patient information is often locked away in silos, inaccessible precisely when it is needed most.

Siloed Patient Records

Millions of patient records remain scattered across hospitals, clinics, pharmacies, and specialist offices, with no standardized mechanism for secure, real-time sharing. A patient moving between providers or jurisdictions may find that their complete health history simply does not follow them.

Inconsistent Data Formats

Without agreed-upon data models and coding standards, the same clinical concept — a diagnosis, a medication, a lab result — may be recorded in dozens of incompatible ways. This inconsistency makes automated data exchange unreliable and error-prone, requiring expensive manual reconciliation.

Patient Safety Risks

Incomplete or inaccessible patient histories contribute directly to adverse events. Clinicians making decisions without full context risk prescribing contraindicated medications, ordering duplicate diagnostic tests, or missing critical allergy information — all preventable with robust data sharing.

Economic Inefficiencies

The Canadian Institute for Health Information (CIHI) and other bodies have estimated that fragmented systems lead to billions of dollars in duplicated diagnostic tests, redundant administrative processes, and avoidable hospitalizations annually. Interoperability is not just a clinical imperative — it is a fiscal one.

Canada's Foundation: The Pan-Canadian Health Data Content Framework (pCHDCF)

A National Blueprint for Health Data Standardization

The Pan-Canadian Health Data Content Framework (PCHDC Framework) represents a landmark collaborative effort between Canada Health Infoway, the Canadian Institute for Health Information (CIHI), provincial and territorial health authorities, and clinical stakeholders. It provides a comprehensive, authoritative blueprint that defines what health data should be collected, how it should be structured, and how it should be shared across the Canadian healthcare continuum.

The Framework is not a technology mandate — it is a conceptual foundation. It establishes the shared vocabulary, data categories, and semantic standards that enable different systems and jurisdictions to "speak the same language" when exchanging health information. By anchoring national efforts to a common content model, the Framework reduces ambiguity and creates the conditions for genuine, scalable interoperability.

Core Objectives

→ Consistency

Ensure health data is recorded and communicated in a standardized, predictable manner across all provinces, territories, and care settings.

→ Accuracy

Define data elements with sufficient precision to eliminate ambiguity and reduce the risk of misinterpretation between sending and receiving systems.

→ Shareability

Make it technically and semantically straightforward to move patient data between authorized systems, enabling coordinated, longitudinal care.

→ Better Outcomes

Ultimately, support safer, more efficient, and more patient-centered healthcare delivery across the country.

Canadian Core Data for Interoperability (CACDI)

The Canadian Core Data for Interoperability (CACDI) is one of the most operationally significant outputs of the Pan-Canadian Health Data Content Framework. It defines the **minimum standardized dataset** that must be exchangeable between health information systems and across jurisdictional boundaries — translating the Framework's conceptual goals into concrete, implementable specifications.

What CACDI Defines

CACDI specifies the precise data elements — patient demographics, allergies, medications, diagnoses, immunizations, clinical notes, and more — that constitute the core of a patient's portable health record. Each element is defined with clear semantics, value sets, and cardinality requirements to ensure that any conformant system can produce and consume the data reliably.

Draft Version 2 of CACDI represents a significant evolution, introducing new data categories identified as priorities by clinical and technical working groups, as well as refining existing elements based on real-world implementation feedback from early adopters across Canadian jurisdictions.

Relationship to CA Core+ (FHIR)

CACDI works in close tandem with Canada Health Infoway's **CA Core+** initiative — a set of pan-Canadian HL7 FHIR profiles that provide the technical expression of CACDI's content requirements. Where CACDI defines *what* data is needed, CA Core+ defines *how* that data should be structured and exchanged using the FHIR standard. Together, they form an integrated content-and-technical specification that implementation teams can use to build conformant, interoperable systems.

Draft Version 2 of CACDI introduces new data categories and refined elements based on pan-Canadian clinical and technical stakeholder input, reflecting the evolving needs of Canada's digital health ecosystem.

The Global Standard: Health Level Seven (HL7) FHIR

Fast Healthcare Interoperability Resources — universally known as **FHIR** (pronounced "fire") — is the next-generation interoperability standard developed and maintained by Health Level Seven International (HL7). Since its initial publication, FHIR has rapidly become the dominant global framework for health data exchange, adopted by regulators, governments, and health systems across North America, Europe, Australia, and beyond.

Modular "Resources"

FHIR is built on discrete, composable data units called "resources" — each representing a specific clinical or administrative concept such as a Patient, Observation, Medication, or Encounter. This modularity allows implementers to adopt only the resources relevant to their use case, reducing complexity and accelerating deployment.

Modern Web Technologies

FHIR leverages familiar, widely supported web standards — RESTful APIs, JSON, XML, and OAuth 2.0 — making it far more accessible to developers than legacy health IT standards. This dramatically lowers the barrier to entry for new implementers and enables integration with consumer-facing applications and mobile devices.

Scalability & Flexibility

FHIR's profiling mechanism allows national and regional bodies to constrain or extend the base standard to meet local requirements — without breaking interoperability with the global community. This balance between a universal foundation and local adaptability is central to FHIR's worldwide success.

Universal Applicability

FHIR is designed to support data exchange across an extraordinary range of contexts: EHR-to-EHR communication, patient-facing apps, public health reporting, clinical decision support, payer-provider transactions, and research data sharing — all using a single, coherent standard.

Canada's FHIR Adoption: CA Core+ and FHIR Exchange (CA:FeX)

Canada has embraced HL7 FHIR as the technical backbone of its national interoperability strategy, with Canada Health Infoway leading the development of two complementary and mutually reinforcing specifications: **CA Core+** and **CA:FeX**. Together, these initiatives translate the pan-Canadian content vision into a fully operational, standards-based exchange ecosystem.

CA Core+: Pan-Canadian FHIR Profiles

CA Core+ is Canada's national set of HL7 FHIR implementation guides and profiles. It takes the international FHIR base standard and applies Canadian-specific constraints — aligning data elements with CACDI requirements, incorporating Canadian coding systems such as SNOMED CT CA and pan-Canadian drug terminologies, and reflecting the governance structures of Canada's federated health system.

CA Core+ profiles define precisely how common clinical resources — Patient demographics, Conditions, Medications, AllergyIntolerance, Immunizations, and more — must be structured when exchanged between conformant Canadian systems. Jurisdictions and vendors implementing CA Core+ can be confident that their data will be interpretable by any other conformant system, regardless of province or technology platform.

CA:FeX: FHIR Exchange Canada

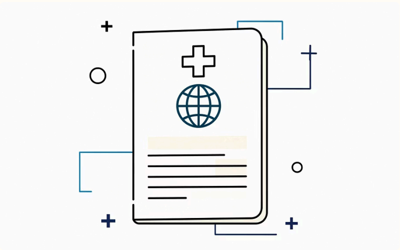
The FHIR Exchange Canada (CA:FeX) specification addresses the *how* of data exchange — the transport, security, and workflow patterns that govern how FHIR-formatted data moves between systems. CA:FeX leverages internationally recognized exchange frameworks, most notably the **International Patient Summary (IPS)** and the **International Patient Access (IPA)** specifications, adapting them for the Canadian context.

The **Patient Summary - Canada (PS-CA)** profile, built on IPS/IPA foundations, defines the content and structure of a standardized patient summary document that can accompany a patient across care settings and provincial borders — ensuring that the most critical clinical information is always available at the point of care.

CA:FeX and CA Core+ are designed to evolve in lockstep with the international FHIR community, ensuring Canada remains aligned with global best practices while meeting domestic requirements.

IPS, IPA, and PS-CA: The International–Canadian Bridge

Three interlocking specifications — the International Patient Summary (IPS), the International Patient Access (IPA), and the Patient Summary Canada (PS-CA) — form a critical bridge between global interoperability standards and Canada's national implementation. Understanding their distinct roles and interdependencies is essential to grasping how Canadian health data exchange will function in practice.



International Patient Summary (IPS)

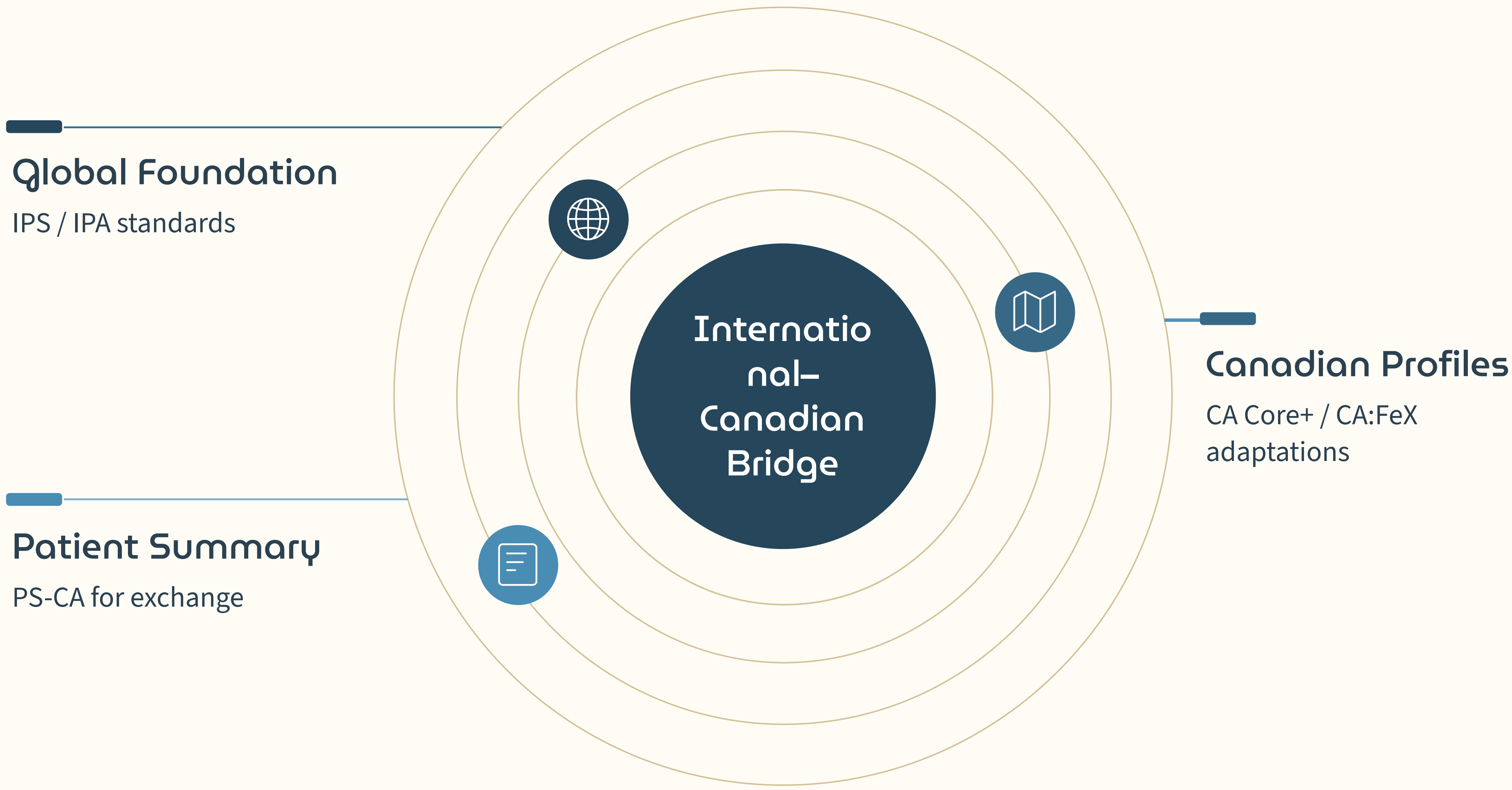
Developed jointly by HL7 and ISO (ISO 27269), the IPS defines the **minimum dataset** required to support safe, unscheduled, cross-border care for an individual patient anywhere in the world. It specifies the clinical sections — allergies, medications, diagnoses, procedures, immunizations, results — that constitute a universally interpretable patient summary. The IPS is now an ISO standard, giving it authoritative global standing.

International Patient Access (IPA)

Where IPS defines content, **IPA** defines access — how patients and authorized applications can query and retrieve their health data using FHIR RESTful APIs. IPA establishes a minimum set of FHIR capabilities that conformant servers must support, enabling patient-facing applications (such as personal health apps) to connect to any IPA-compliant health system and retrieve a consistent, meaningful dataset.

Patient Summary — Canada (PS-CA)

PS-CA is Canada's national instantiation of the IPS, developed through CA:FeX as a FHIR Implementation Guide. It takes the IPS content model and applies Canadian-specific requirements: Canadian drug terminology, SNOMED CT CA, provincial identifier systems, and governance rules aligned with Canadian privacy legislation (PIPEDA and provincial equivalents). PS-CA ensures that a patient summary generated in British Columbia is fully interpretable in Ontario or Nova Scotia — and, through IPS alignment, potentially in any country that has adopted the international standard.



This layered architecture — global standard, national profile, patient-facing document — ensures that Canada's interoperability investments are both locally relevant and internationally compatible, positioning the country as an active participant in the global movement toward seamless health data exchange.

Canadian Health Application Lightweight Protocol (HALO)

The Canadian Health Application Lightweight Protocol (HALO) is a pivotal pan-Canadian interoperability framework developed by **Canada Health Infoway**. Its core purpose is to facilitate the seamless integration of third-party digital health applications with existing clinical systems, such as Electronic Medical Records (EMRs).

Seamless Integration

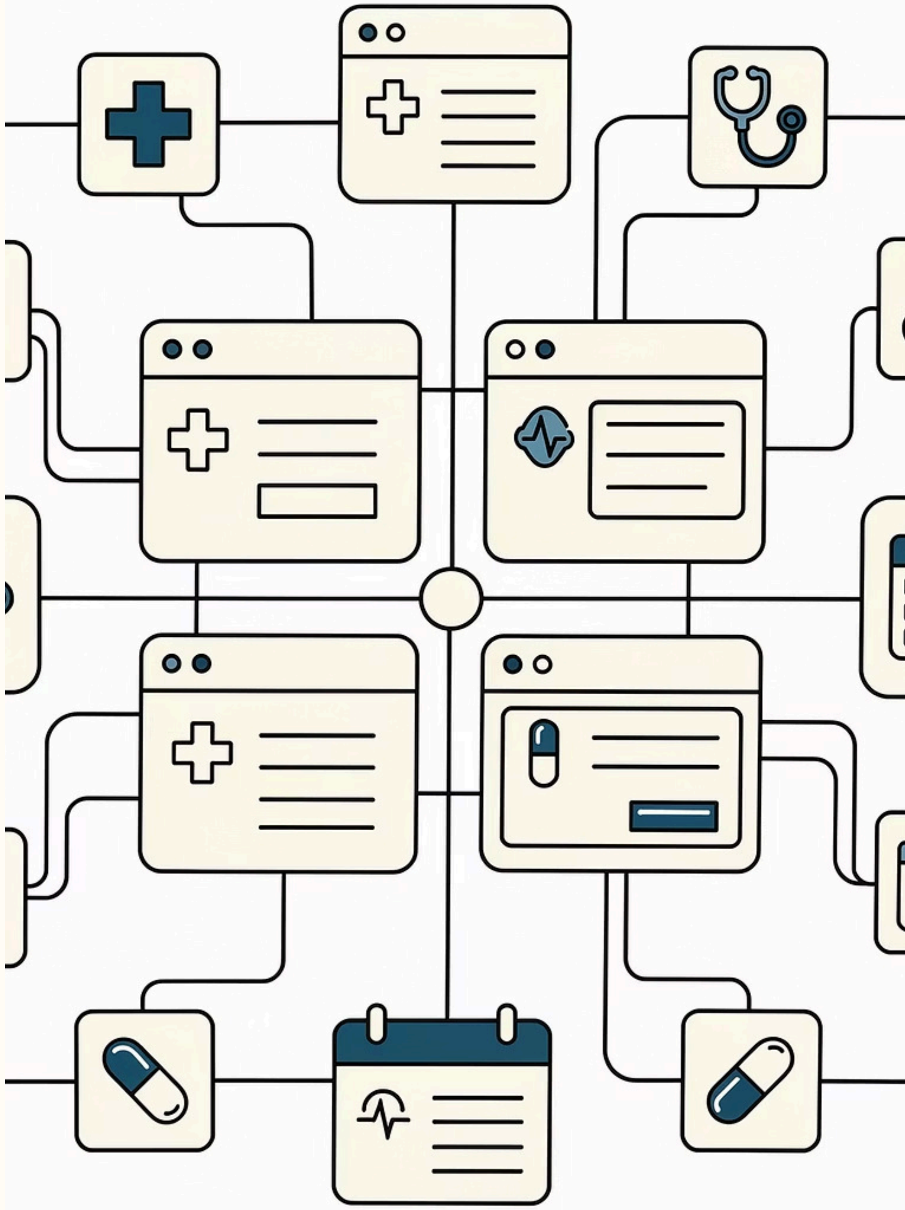
HALO establishes a standardized pathway for external digital health applications to connect directly with EMRs, enabling a unified and efficient data exchange experience for healthcare providers.

"Plug and Play" Capabilities

This framework is designed for ease of use, allowing new applications to be integrated with minimal effort. This "plug and play" approach accelerates the deployment of innovative health solutions.

Enhanced Clinician Workflow

By integrating tools like eReferrals and patient data viewers directly into EMRs, HALO eliminates the need for multiple logins and disparate systems, significantly streamlining clinical workflows and improving efficiency at the point of care.



HALO & SoFA: FHIR Subscriptions and Real-Time Data Sync

Within HALO, the **SMART on FHIR Accelerator (SoFA)** enables Point of Care (PoC) systems that lack a native FHIR API or SMART-compliant authorization server to participate in the SMART App Launch flow. SoFA uses the **$set-context** operation to populate FHIR resources and clinical context before launching a SMART application — but to handle data flowing *back* to the PoC, HALO defines **FHIR Subscriptions** as the primary synchronization mechanism.

1

SoFA Content Update Topic

Canonical URL: `http://fhir.infoway-inforoute.ca/io/HALO/SubscriptionTopic/sofa-content-update`. PoC systems must maintain an active Subscription to this topic *before* initiating any SMART app launch. SoFA validates this during **$set-context** processing and rejects the operation if no active Subscription exists.

2

FHIR R5 Subscriptions Backport

HALO is built on FHIR R4 for CA Core+ compatibility, but adopts the **FHIR R5 Subscriptions Backport IG** to enable a modern, topic-based, event-driven notification model — without requiring a full migration to FHIR R5.

3

Subscription Channels: REST-Hook & WebSocket

HALO constrains supported channels to `rest-hook` (HTTP/S POST to a PoC endpoint) and `websocket` (persistent connection via a `$get-ws-binding-token` binding). Both channels include a handshake verification step before the Subscription becomes `active`.

4

Event Tracking & Recovery

Each notification includes a sequential `eventNumber`. PoC systems must track these to detect gaps and use the `$events` operation to retrieve missed notifications — including on startup, after errors, or when gaps are detected in the event sequence.

SoFA's subscription model ensures PoC systems are not passive context providers — they continuously synchronize with clinical data updates generated by SMART applications, maintaining data integrity across the care ecosystem.

pan-Canadian Care Service Directory (CA-CSD)

The pan-Canadian Care Service Directory (CA-CSD) is a national FHIR Implementation Guide developed to standardize the way care services and organizations expose their information. It enables efficient discovery, search, and referral for healthcare providers and patients across Canada, directly addressing fragmentation in health service access.

Seamless Discovery

Provides a standardized mechanism for finding healthcare services, organizations, and practitioners, including detailed information like specialties, hours, eligibility criteria, and wait times.

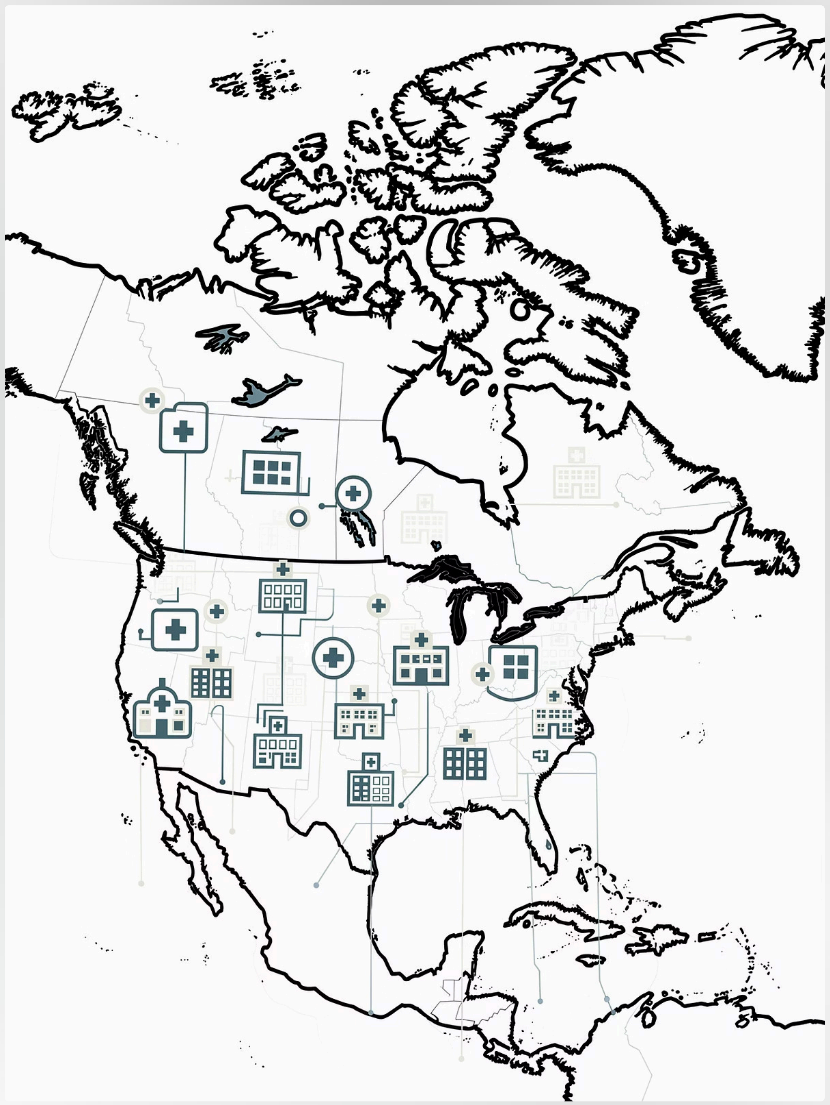
Standardized Referrals

Leverages FHIR ServiceRequest and CareTeam resources to facilitate consistent and trackable digital referrals between different care settings, improving coordination of care.

Enhanced Access & Interoperability

By defining a common language for service information, CA-CSD helps to bridge gaps, ensuring that all Canadians can more easily access the right care at the right time, fostering a truly pan-Canadian health ecosystem.

This initiative is crucial for building a more integrated and patient-centric healthcare system, allowing for better navigation and utilization of available health resources.



Important Disclaimer

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